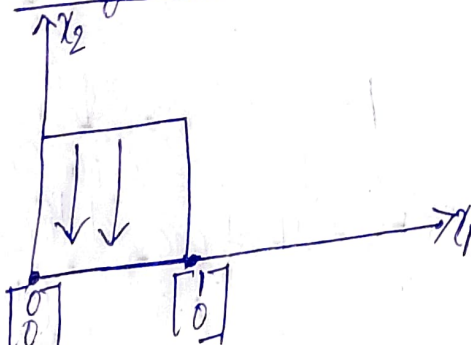


Table 4 Projections

Transformation

Projection onto the x_1 -axis

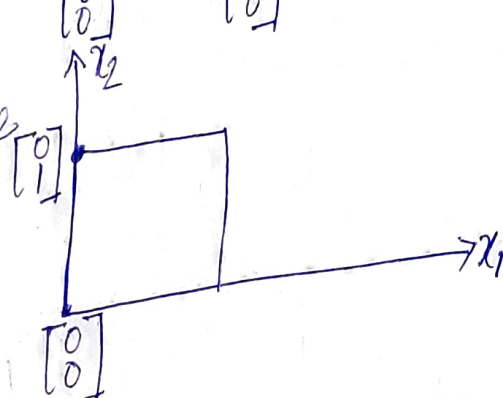
Image of the Unit Square



Standard Matrix

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

Projection onto the x_2 -axis



$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Existence and Uniqueness Questions

Definition: A mapping $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be **onto** $\mathbb{R}^m$ if each b in $\mathbb{R}^m$ is the image of at least one x in $\mathbb{R}^n$.

Equivalently, T is onto $\mathbb{R}^m$ when the range of T is all of the codomain $\mathbb{R}^m$. That is, T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if, for each b in the codomain $\mathbb{R}^m$, there exists at least one solution of $T(x) = b$.

"Does T map $\mathbb{R}^n$ onto $\mathbb{R}^m$?" is an existence question. which
The mapping T is not onto when there is some b in $\mathbb{R}^m$ for the equation $T(x) = b$ has no solution.

Definition: A mapping $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be **one-to-one** if each b in $\mathbb{R}^m$ is the image of at most one x in $\mathbb{R}^n$.

Equivalently, T is one-to-one if, for each b in $\mathbb{R}^m$, the equation $T(x) = b$ has either a unique solution or none at all.
"Is T one-to-one?" is a uniqueness question.
The mapping T is not one-to-one when some b in $\mathbb{R}^m$ is the image of more than one vector in $\mathbb{R}^n$. If there is no such b , then T is one-to-one.